

Silicon Valley (in the US), Cambridge (in the UK) and Taiwan, in terms of chip designing clusters.

India is also emerging as a prominent player in wireless IC design with about 38 per cent of Indian designers working on designs targeting wireless and mobile applications compared to just about 22 per cent in Europe.

Why India?

Traditionally, Taiwan has been the favourite destination for chip designing, but during the last few years, India has also seen the market grow. The availability of a large engineering talent pool and cost advantages as compared to other countries have been the key growth drivers for this industry in India.

Sanjay Mittal, managing director of Yogasa Systems, mentions that some of the other influencing factors include a strong technical education system, reduced entry barriers, government support in the form of STPI and EOU schemes, and a strong intellectual property rights (IPRs) framework.

He adds, "India is at the forefront of VLSI design owing to the cutting-edge chip design activities taking place in multinational design companies that have large-scale engineering operations here. These captive units take advantage of the availability of skilled and low-cost workforce to develop products for global markets. If the same is to be done in their respective countries, the development cost will be at least 300 per cent higher than in India."

Factors that make India an attractive destination

- Highly skilled and cost-effective workforce
- Respect for intellectual property rights (IPRs): India has a good track record of safeguarding the IPRs of foreign companies, which encourages them to establish design centres in India
- Emergence of R&D and design centres of homegrown companies, including software providers like Wipro and TCS
- Emergence of start-up design firms
- A rise in outsourcing: Asia is the hub for electronics manufacturing services, primarily because of low manufacturing cost and relatively low-cost labour, and pressure from consumer electronics/telecommunications industry

Challenges that need to be addressed

- Need to provide end-to-end services
- Poor infrastructure facilities
- High investment costs for R&D
- Competition from other Asian countries
- Inadequate support from government for the capital-intensive semiconductor industry
- Talent shortage
- Demand-supply gap for chip design engineers

Up to the value chain

India has been quick to tap into this opportunity. The semiconductor industry in India has grown rapidly in the last three to five years with the establishment of captive design centres of US chip companies and the evolution of IP-leveraged design services firms. Well-known IT services firms have also entered the market with the setting up of separate practices for electronics design services. Local companies and design subsidiaries are also evolving from the 'build-to-spec' model.

Anil Kumar, managing director of SLN Technologies, observes a discernible transition among many captive

design centres from being engineering subsidiaries to becoming product companies.

"Increasingly, companies are realising that cost arbitrage will take them only to a certain level, and thus they are seeking avenues for 'value' creation with innovative designs. It may create good opportunity for Indian indigenous design service providers," says Rajiv Batra, managing director of Rabyte.

As of today, the companies in India are involved in various phases of the product design cycle. There are a number of companies which have been in this area for a long time to have built capabilities to move up the value chain. Some are trying to tie-up with architecture providers such as ARM, Intel, Marvell, Maxim, NXP and Texas Instruments and then develop products based on their architectures. These companies are getting to a level where, right from using the architecture, they can start with specifications to rolling out the final product.

Global slowdown, rising opportunities

Globally, the design industry and the worldwide market for semiconductor

The potential for growth is huge for Indian design service providers. The top five end-user products that are expected to drive growth would be mobile handsets, desktops and notebooks, GSM base stations, set-top boxes and energy meters.

—Partha Pratim Roy, Advisor, Global Consumer Lab, Ericsson-Market Unit India & Sri Lanka